

From FQHE to Fuzzy Sphere Numerical CFT

Landau Levels as a Regulator for 3D Criticality

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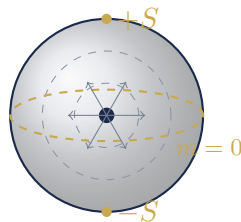
Outline

1. Starting Point: Haldane Sphere
2. LLL Projection $\rightarrow$ Fuzzy Geometry
3. Beyond FQHE
4. Hamiltonian Construction
5. State-Operator Correspondence
6. Numerical Workflow
7. Why Not a Lattice?
8. Conclusion

Haldane Sphere in FQHE

- Particles on S^2 with magnetic monopole flux $2S$
- Lowest Landau level: $\dim \mathcal{H}_{\text{LLL}} = 2S + 1$
- Single-particle basis forms spin- S rep of $\text{SO}(3)$

$$m = -S, \dots, S \quad \dim = N_\phi + 1 = 2S + 1 \quad (1)$$



Key message

The sphere gives a **finite, rotationally symmetric** Hilbert space.

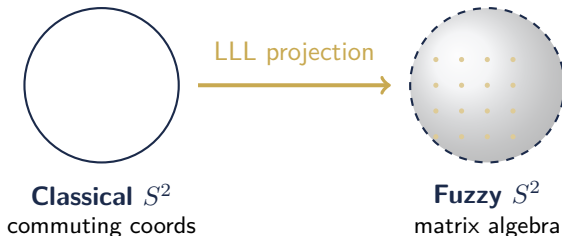
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LLL Projection $\rightarrow$ Fuzzy Sphere

- After LLL projection, kinetic energy is **quenched**
- Coordinates no longer commute under projection
- Finite matrix algebra approximates functions on S^2

$$[X_i, X_j] \sim \frac{i}{\sqrt{S(S+1)}} \epsilon_{ijk} X_k \quad (2)$$



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Why This Matters Beyond FQHE

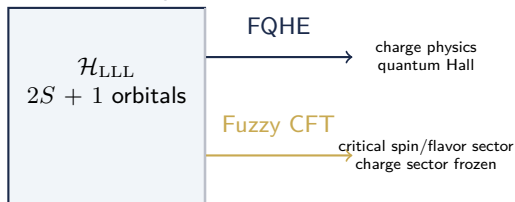
Traditional FQHE use

- Study charge/topological quantum Hall physics
- Interactions *inside* LLL determine phase

Fuzzy sphere CFT use

- Same LLL Hilbert space → **UV regulator**
- Internal flavors encode target field theory

LLL Hilbert Space



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Constructing a Fuzzy-Sphere Hamiltonian

- Choose LLL creation operators $c_{m,a}^\dagger$
- m = orbital index, a = flavor/spin
- Write **rotationally invariant** interactions projected to LLL

$$H = \sum_{\ell} V_{\ell} P_{\ell} + \text{tuning terms} \quad (3)$$

$$H = P_{\text{LLL}} H_{\text{int}} P_{\text{LLL}} \quad (4)$$

Not: put a lattice CFT Hamiltonian on a sphere
Instead: build a many-body Hamiltonian
 whose **IR fixed point is the target CFT**

Key message

Tune coupling constants to reach a **critical point**.

Outline

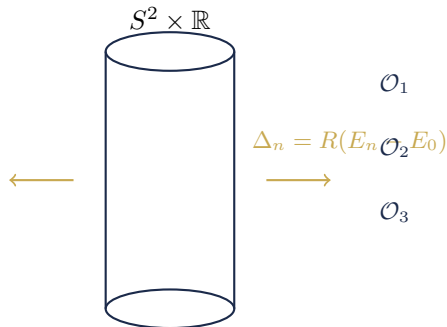
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State-Operator Correspondence on $S^2 \times \mathbb{R}$

- For 2 + 1D CFT on S^2 , energy levels $\leftrightarrow$ scaling dimensions
- Angular momentum labels $\text{SO}(3)$ spin of operators

$$E_n - E_0 = \frac{\Delta_n}{R} \quad (5)$$

$$\text{states on } S^2 \longleftrightarrow \text{local operators in } \mathbb{R}^3 \quad (6)$$



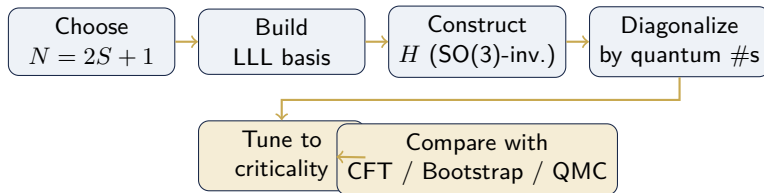
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Exact diagonalization gives an **operator spectrum**, not just a phase diagram.

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Numerical Workflow



Key observables

- Energy spectrum
- Angular momentum sectors
- Scaling dimensions Δ_n

Advanced

- OPE coefficients
- Correlation functions
- Finite-size scaling

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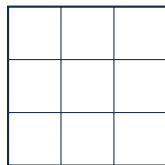
Why Not Just Use a Lattice?

Lattice approach

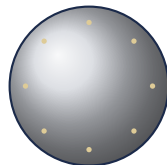
- Breaks continuous rotation
- Strong microscopic anisotropy
- CFT data muddled by irrelevant operators
- Cubic symmetry $\neq$ full $SO(3)$

Fuzzy sphere approach

- $SO(3)$ exact at *any* finite size
- CFT data organized by angular momentum
- Finite-size scaling $\sim$ radial quantization
- No lattice artifacts



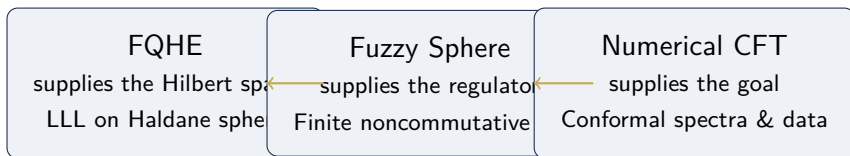
better



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Take-Home Message



The surprise:
A tool born in quantum Hall physics
becomes a **microscope for 3D conformal field theory**

Thank You

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